

Pneumonia Detection from Chest X-Ray Images

Using a CNN-Based Architecture (ResNet-18 with Transfer Learning)

Medical Image Analysis

Item	Details
Name	Md. Abdullah Al Mamun
Institute	BRAC University
Task	Medical image classification (binary: NORMAL vs. PNEUMONIA)
Dataset	Chest X-Ray Images (Pneumonia) – Kermany et al., via Kaggle
Model	ResNet-18 (ImageNet-pretrained) with a custom classification head
Framework	PyTorch / torchvision, scikit-learn (Google Colab, CPU runtime)
GitHub Repository	https://github.com/aam-bd/medical-chest-xray-pneumonia-cnn.git

1. Introduction and Objective

Pneumonia is a lung infection that is commonly diagnosed with the help of chest radiographs. Reading these images requires expert radiologists, who may not be available everywhere, so automated decision-support tools built with deep learning are an active area of research.

This project builds an image classification model based on a Convolutional Neural Network (CNN) that labels a chest X-ray as either NORMAL or PNEUMONIA. The work covers the complete pipeline:

- loading a public medical imaging dataset and performing exploratory data analysis;
- preprocessing, augmentation and a train / validation / test split;
- implementing and training a CNN (ResNet-18 fine-tuned by transfer learning);
- evaluating the model on a held-out test set with suitable metrics (accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC);
- visualising training curves and sample predictions.

2. Dataset and Preprocessing

2.1 Dataset description

The dataset is the publicly available Chest X-Ray Images (Pneumonia) collection by Kermany et al., downloaded from Kaggle (paultimothymooney/chest-xray-pneumonia) using the kagglehub library. It contains paediatric chest radiographs stored as JPEG files in two classes, NORMAL and PNEUMONIA, and is distributed in three folders (train, val and test). The number of images per folder is shown in Table 1.

Original folder	NORMAL	PNEUMONIA	Total
train	1,341	3,875	5,216
val	8	8	16
test	234	390	624
All	1,583	4,273	5,856

Table 1. Image counts in the original dataset folders.

2.2 Exploratory data analysis

Figure 1 shows the class distribution of the training folder. The data are clearly imbalanced: there are about 2.9 times as many PNEUMONIA images (3,875) as NORMAL images (1,341). Without correction, a model could achieve high accuracy simply by favouring the majority class, so this imbalance is handled explicitly during training (Section 2.4).

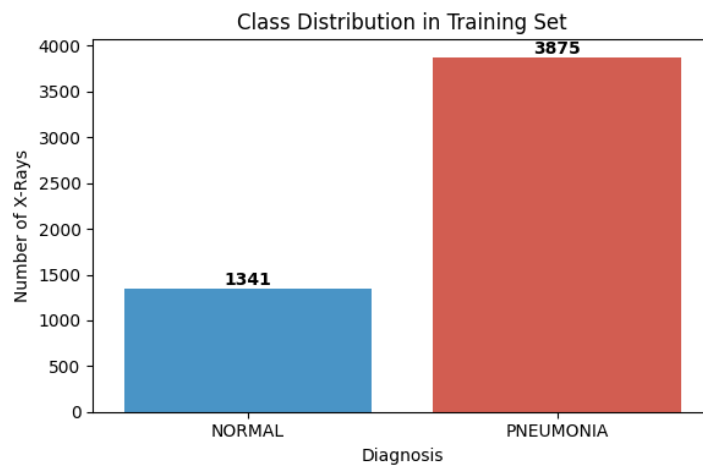


Figure 1. Class distribution in the training set.

Figure 2 shows four random images from each class. The radiographs differ in image size and aspect ratio, exposure and contrast, patient positioning, and contain text markers (for example the "R" side label) and, in some cases, medical devices such as leads and tubes. PNEUMONIA images tend to show hazier, more opaque lung fields, but the visual difference from normal images is often subtle, which motivates the use of a CNN that can learn discriminative texture and shape features automatically.

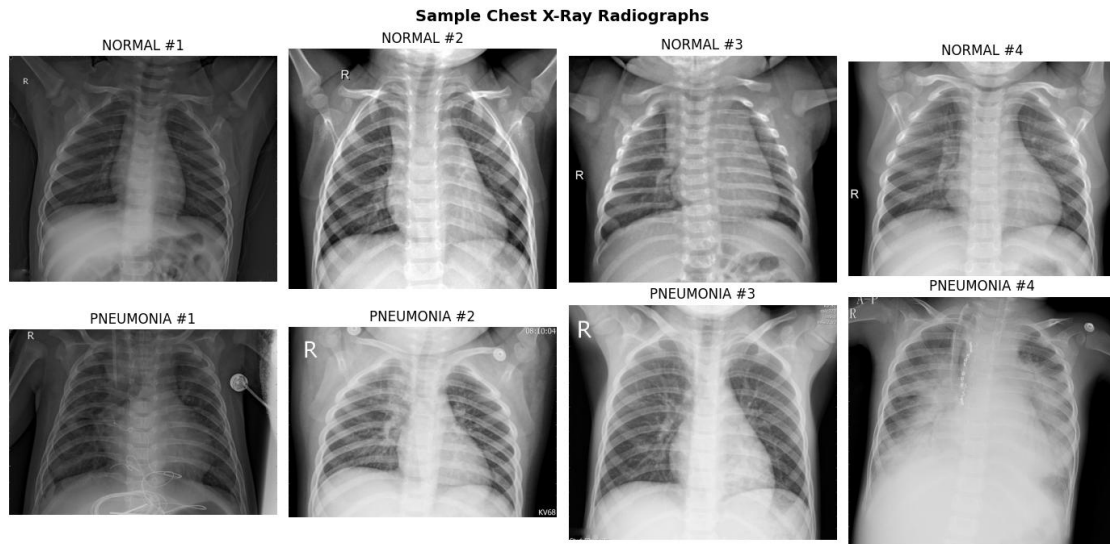


Figure 2. Sample chest X-ray images from the NORMAL (top row) and PNEUMONIA (bottom row) classes.

2.3 Preprocessing and data augmentation

All images are converted to three-channel RGB so that they match the input format of the ImageNet-pretrained backbone, then processed as follows:

- **Resizing:** every image is resized to 224×224 pixels.
- **Normalisation:** pixel values are scaled to $[0, 1]$ and standardised with the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).
- **Augmentation (training data only):** random horizontal flip ($p = 0.5$), random rotation of up to $\pm 10^\circ$, and colour jitter (brightness and contrast ± 0.1). Validation and test images receive only resizing and normalisation, so evaluation is deterministic.

2.4 Train / validation / test split and class imbalance

The official validation folder contains only 16 images, which is too small for reliable model selection. It was therefore merged with the training folder (5,232 images) and re-split randomly into 85% training and 15% validation using a fixed random seed (42) for reproducibility. The official test folder (624 images) was left untouched and was used only once, for the final evaluation.

Subset	Images	Source / share	Purpose
Training	4,448	85% of train+val pool	Weight updates
Validation	784	15% of train+val pool	Model selection / LR schedule
Test	624	Official test folder	Final evaluation only

Table 2. Final data split used in the notebook.

To counter class imbalance, the loss function uses a positive-class weight equal to the ratio of NORMAL to PNEUMONIA images in the training pool ($1,349 / 3,883 = 0.3474$). This down-weights the majority PNEUMONIA class so that both classes contribute more equally to the gradient.

3. CNN Architecture and Methodology

3.1 Choice of architecture

The model is built on ResNet-18, a residual CNN with 18 weight layers. Each residual block (BasicBlock) contains two 3×3 convolutions with batch normalisation and ReLU, plus a skip connection that adds the block input to its output. The skip connections make deep networks easier to optimise and reduce the vanishing-gradient problem. ResNet-18 is small enough to train on modest hardware (this notebook ran on a CPU runtime) yet strong enough for medical imaging when initialised from ImageNet weights.

Because the medical dataset is relatively small, transfer learning is used: the network starts from ImageNet-pretrained weights, which already encode general edges, textures and shapes, and only the deepest layers plus a new classifier head are adapted to chest X-rays.

3.2 Model structure

Stage	Layers	Output (C × H × W)	Training
Input	RGB X-ray, 224×224	$3 \times 224 \times 224$	—
conv1	7×7 conv, 64, stride 2 + BN + ReLU	$64 \times 112 \times 112$	Frozen
maxpool	3×3 max-pool, stride 2	$64 \times 56 \times 56$	Frozen
layer1	2 × BasicBlock (64 channels)	$64 \times 56 \times 56$	Frozen
layer2	2 × BasicBlock (128 channels)	$128 \times 28 \times 28$	Frozen
layer3	2 × BasicBlock (256 channels)	$256 \times 14 \times 14$	Frozen
layer4	2 × BasicBlock (512 channels)	$512 \times 7 \times 7$	Partly fine-tuned
Global avg-pool	Adaptive average pooling	512	No parameters
Custom head	Dropout(0.4) → Linear(512→128) → ReLU → Dropout(0.2) → Linear(128→1)	1 (logit)	Trained from scratch

Table 3. Architecture of the pneumonia classifier.

The original 1000-class ImageNet layer is replaced by a small two-layer head. The two dropout layers (0.4 and 0.2) regularise the head, and the single output neuron produces one logit; a sigmoid converts it to the probability of PNEUMONIA, and a threshold of 0.5 gives the predicted class.

3.3 Transfer-learning strategy

All parameter tensors except the last 12 of the pretrained backbone are frozen. In effect, conv1 and layer1–layer3 keep their ImageNet weights, while the last residual block (layer4, block 2), the shortcut projection of layer4 block 1 and the final batch-norm bias are fine-tuned, along with the entire new classifier head. Approximate parameter counts, derived from the architecture, are given in Table 4.

Parameters	Count	Share
Trainable (fine-tuned layer4 parts + new head)	≈ 4.92 million	≈ 44%
Frozen (conv1, layer1–3, part of layer4)	≈ 6.32 million	≈ 56%
Total	≈ 11.24 million	100%

Table 4. Approximate trainable and frozen parameters.

Note that the model is placed in training mode during training, so the batch-normalisation running statistics of the frozen layers continue to adapt to the X-ray data even though their weights are not updated.

3.4 Training configuration

Setting	Value
Loss function	BCEWithLogitsLoss with pos_weight = 0.3474 (handles class imbalance)
Optimiser	AdamW, learning rate 1e-4, weight decay 1e-3 (trainable parameters only)
LR scheduler	ReduceLROnPlateau on validation loss (factor 0.5, patience 2)
Batch size / epochs	32 / 10
Model selection	Checkpoint saved whenever validation loss improves (best_pneumonia_cnn.pth)
Random seed	42 (PyTorch, CUDA and NumPy)
Hardware	Google Colab, CPU runtime

Table 5. Hyper-parameters and training setup.

4. Training Process and Results

In each epoch the model is trained on the augmented training set and then evaluated on the validation set. Loss and accuracy are recorded for both, the learning-rate scheduler is stepped with the validation loss, and the weights are saved whenever the validation loss reaches a new minimum. After training, the best checkpoint (not the last epoch) is reloaded for testing.

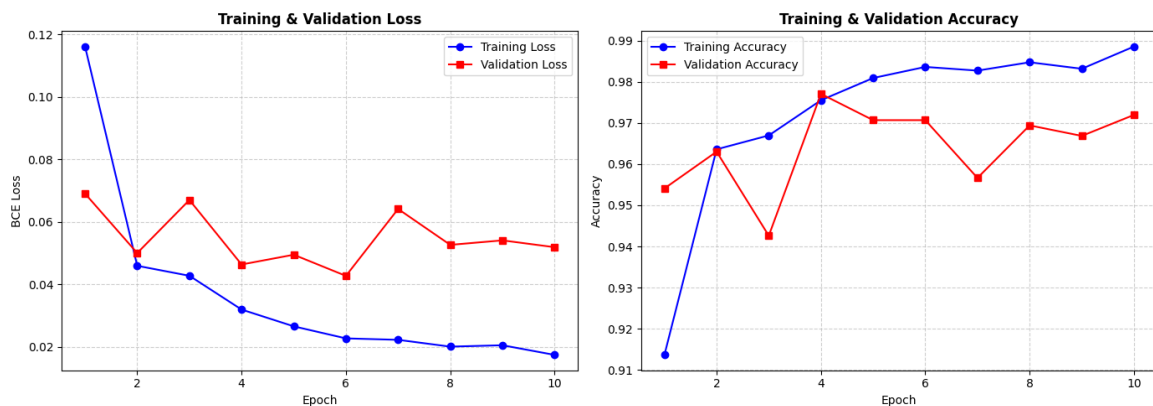


Figure 3. Training and validation loss (left) and accuracy (right) over 10 epochs.

Epoch	Train loss	Train acc.	Val loss	Val acc.
1	0.1161	91.37%	0.0690	95.41%
2	0.0459	96.36%	0.0500	96.30%
3	0.0427	96.70%	0.0670	94.26%
4	0.0319	97.55%	0.0463	97.70%
5	0.0265	98.09%	0.0495	97.07%
6	0.0227	98.36%	0.0427	97.07%
7	0.0222	98.27%	0.0641	95.66%
8	0.0200	98.47%	0.0526	96.94%
9	0.0205	98.31%	0.0541	96.68%
10	0.0174	98.85%	0.0519	97.19%

Table 6. Per-epoch results. The highlighted row (epoch 6) has the lowest validation loss and was saved as the final model.

Key observations:

- Learning is fast: training accuracy rises from 91.4% to 98.9% and training loss falls steadily from 0.116 to 0.017.
- Validation accuracy stays between roughly 94% and 97.7%. The lowest validation loss (0.0427) occurs at epoch 6, so the epoch-6 weights were kept.
- After epoch 6 the validation loss stops improving while the training loss keeps decreasing, a mild sign of overfitting. Selecting the checkpoint by validation loss guards against this.

5. Test-Set Evaluation

The best checkpoint was evaluated on the 624 held-out test images, which were not used for training or model selection. Results are in Table 7 and Figure 4.

Class	Precision	Recall	F1-score	Support
NORMAL (0)	0.9744	0.6496	0.7795	234
PNEUMONIA (1)	0.8248	0.9897	0.8998	390
Macro average	0.8996	0.8197	0.8396	624
Weighted average	0.8809	0.8622	0.8547	624

Table 7. Classification report on the test set.

Overall metric	Value
Accuracy	86.22% (538 / 624)
ROC-AUC	0.9577
Sensitivity (recall for PNEUMONIA)	98.97% (386 / 390)
Specificity (recall for NORMAL)	64.96% (152 / 234)

Table 8. Summary metrics on the test set.

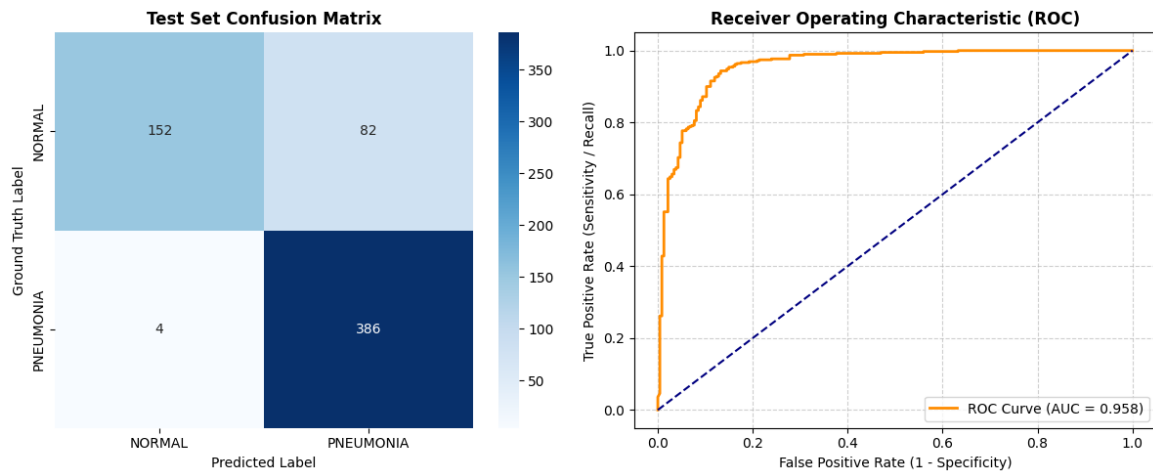


Figure 4. Test-set confusion matrix (left) and ROC curve (right).

Interpretation

- **Very few missed pneumonia cases.** Only 4 of 390 pneumonia X-rays were classified as normal (false negatives). High sensitivity is the priority in a screening setting, where missing a disease is costlier than a false alarm.
- **Many false alarms.** 82 of 234 normal X-rays were predicted as pneumonia (false positives), which is why specificity is only 65% and PNEUMONIA precision is 82%. When the model predicts NORMAL, it is correct 97.4% of the time.
- **Good ranking ability.** The ROC-AUC of 0.958 shows the model separates the two classes well across thresholds. The gap between AUC and accuracy suggests the default 0.5 threshold is not optimal for the NORMAL class.
- **Validation vs. test gap.** Validation accuracy (about 97%) is noticeably higher than test accuracy (86%). Plausible reasons are: (i) the official test images are known to differ somewhat in distribution from the training images in this dataset; (ii) the split is done at image level, so several images from the same patient may fall in both the training and validation sets, making validation optimistic; and (iii) the class-weighted loss together with a 0.5 threshold pushes predictions toward PNEUMONIA. These are hypotheses that were not tested separately in this project.

6. Sample Predictions

Figure 5 shows model predictions on the first eight test images. The title of each image gives the true label, the predicted label and the model confidence; green titles mark correct predictions and red titles mark errors.

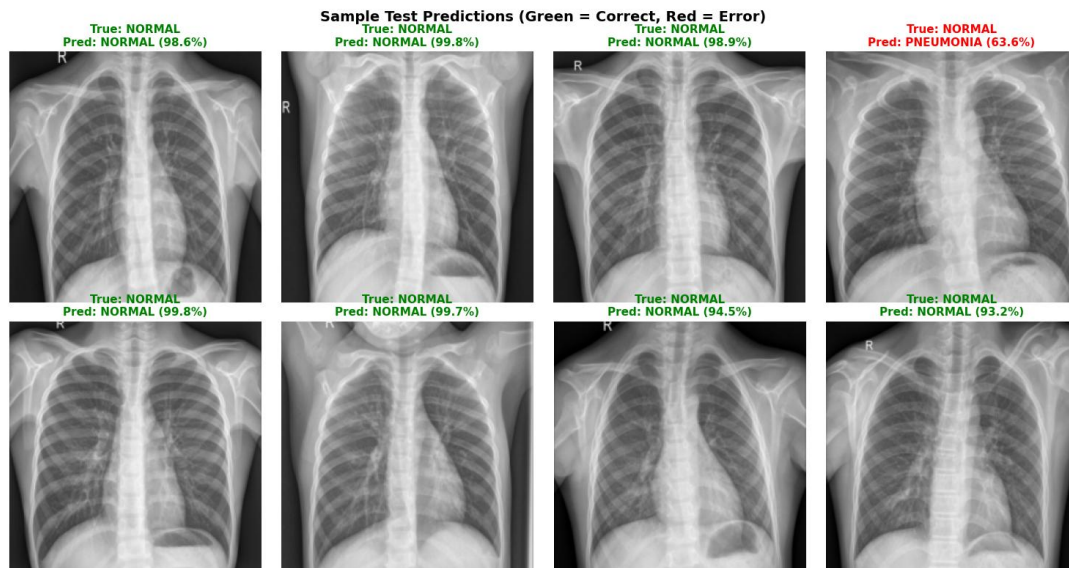


Figure 5. Sample test predictions (green = correct, red = error).

The test loader is not shuffled and the folder lists NORMAL images first, so all eight images shown here happen to be NORMAL. Seven are classified correctly with confidence between 93.2% and 99.8%. The one error (top right) is a normal X-ray predicted as PNEUMONIA with a comparatively low confidence of 63.6%, consistent with the false-positive tendency seen in the confusion matrix. Predictions on PNEUMONIA images are summarized by the confusion matrix (386 of 390 correct).

7. Limitations and Future Work

- Specificity is limited. The decision threshold could be tuned on the validation ROC curve (or a cost-sensitive criterion) to balance sensitivity and specificity.
- Use a patient-level split (or cross-validation) to avoid possible leakage between training and validation images.
- Add explainability, for example Grad-CAM heat maps, to check that the model attends to lung regions rather than image markers, devices or text labels.
- Horizontal flipping changes the side of the heart, which is anatomically unusual for chest X-rays; augmentation choices could be reviewed.
- Train longer, unfreeze more layers, or compare other CNNs (DenseNet, EfficientNet) and ensembles; validate on an external dataset.
- The model is an educational prototype and is not a certified diagnostic tool.

8. Conclusion

A ResNet-18 CNN, initialised from ImageNet weights and fine-tuned on paediatric chest X-rays, reached about 97% validation accuracy and 86.2% accuracy with a ROC-AUC of 0.958 on the held-out test set. The model detects pneumonia very reliably (98.97% sensitivity) but produces a substantial number of false positives (64.96% specificity). The complete pipeline, covering data loading, EDA, preprocessing, class-imbalance handling, training, evaluation and visualisation, are completed, and the main next steps are threshold calibration, patient-level validation and interpretability analysis.

9. Code Repository and Reproducibility

GitHub repository: <https://github.com/aam-bd/medical-chest-xray-pneumonia-cnn.git>

The repository contains the Jupyter Notebook with all code and outputs. To reproduce the results, open the notebook in Google Colab (or a local Jupyter environment with Kaggle access), and run all cells in order. Main libraries: PyTorch, torchvision, scikit-learn, NumPy, Matplotlib, Seaborn, Pillow and kagglehub. Files produced by the notebook: best_pneumonia_cnn.pth (best weights), training_curves.png, evaluation_metrics.png and sample_predictions.png. A fixed random seed (42) is used throughout.

References

- [1] Kermany, D. S., et al. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5), 1122–1131.
- [2] Mooney, P. T. Chest X-Ray Images (Pneumonia). Kaggle dataset: kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.
- [3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. *CVPR*.
- [4] Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularization. *ICLR*.
- [5] Paszke, A., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. *NeurIPS*.
- [6] <https://huggingface.co/datasets/TianyuZhang/ModelMergingBaseline16Datasets/blame/main/datasets/weather.py>
- [7] <https://github.com/ryancodingg/CIFAR-10-image-classification-with-CNN-and-transfer-learning>